A history criterion for the non-axisymmetric part of a viscous flow

September 27, 2026

Abstract

We give a sufficient regularity criterion when the angular mean of a three-dimensional Navier–Stokes velocity satisfies prescribed anisotropic bounds. The additional hypothesis is finiteness of the critical $L^2_t L^\infty_x$ norm of the angular fluctuation. Its Reynolds stress vanishes in the anisotropic ancient limit without differentiating the fluctuation. A circulation barrier for that limit eliminates the swirl; a cylindrical stretching identity then closes the full-velocity enstrophy estimate using small mean derivatives. We also give a separate quantitative gradient-history estimate and exact finite-energy examples of both signs of kinetic transfer between the mean and the fluctuation. The proof adapts the axisymmetric argument of Constantin, Ignatova, and Vicol. This is an analytic conditional criterion, not a Lean theorem or an arbitrary-data regularity proof.

1 Statement

Let $Q = B(1) \times (-1, 0)$ and let (u, p) be a suitable weak solution of

$$u_t + (u \cdot \nabla)u - \Delta u + \nabla p = 0, \qquad \operatorname{div} u = 0,$$

smooth on compact subsets of Q. Assume the usual local energy class $u \in L^\infty_t L^2_x \cap L^2_t H^1_x$ and $p \in L^{3/2}(Q)$. For rotations Q_ϕ about the fixed third axis put

$$\mathcal{P}u(x) = \frac{1}{2\pi} \int_0^{2\pi} Q_\phi u(Q_\phi^{-1} x)\, \mathrm{d}\phi, \qquad v = \mathcal{P}u, \qquad w = u - v.$$

The same notation denotes the induced action on tensors. Write $r = |x'|$, $z = x_3$, and $\tau = -t$. Unless a smaller ball is specified, suprema are on $B(1)$.

Theorem 1. *Suppose $0 < h < 1/2$ and, for every $a + b \le 2$,*

$$|\partial_r^a \partial_z^b v_r| \le C \tau^{-1/2 - a/2 - (1/2-h)b}, \tag{1}$$

$$|\partial_r^a \partial_z^b v_z| + |\partial_r^a \partial_z^b v_\theta| \le C \tau^{-1/2 - h - a/2 - (1/2-h)b}. \tag{2}$$

Set $K(t) = \|w(t)\|_\infty$. If

$$\int_{-1}^0 K(t)^2 \, \mathrm{d}t < \infty, \tag{3}$$

then $(0, 0)$ is a regular point of u.

The mean bounds are those used in [1]. Their validity and (3) are hypotheses, not consequences of the energy inequality. The axis and the ball are fixed; the theorem makes no assertion about propagating approximate symmetry from a shrinking core. No derivative or torque integral of w is assumed. The proof uses the full second-order swirl bounds in (2); it does not extend the meridional-only variant of [1, Remark 5.2] to this setting.

Corollary 2. *In Theorem 1, either of the following bounds near zero suffices in place of (3):*

$$K(t) \le C\tau^{-\beta}, \quad 0 \le \beta < \tfrac{1}{2}; \qquad K(t) \le C\tau^{-1/2}\log(e/\tau)^{-p}, \quad p > \tfrac{1}{2}.$$

These implications follow by integrating K^2 and restricting to a final cylinder. In particular no gradient estimate is inferred from an amplitude bound. A stronger weighted amplitude condition does pay a gradient history in the whole-space setting; that independent estimate is in Section 6.

2 The mean equation and its rescaled stress

Rotation covariance and $\mathcal{P}w = 0$ give the exact equation

$$v_t + (v\cdot\nabla)v - \Delta v + \nabla\mathcal{P}p = F, \qquad F = -\mathcal{P}[(w\cdot\nabla)w] = -\operatorname{div}\mathcal{P}(w\otimes w). \tag{4}$$

The proof retains this divergence form and does not require an L^∞ bound on F.

The original solution has a regular spherical annulus inside any prescribed interval of radii. Indeed, partial regularity makes the top-slice singular set have zero $\mathcal{H}^1$ measure. Its image under $x \mapsto |x|$ has measure zero, so some intervening sphere is regular. Compactness and interior estimates give an annulus on which u, ∇u, and $\nabla^2 u$ are bounded through the endpoint, as in the first part of [1, Lemma 3.2]. Its rotational invariance gives the same bounds for v and w. This argument uses suitability of u; it does not require a partial-regularity theorem for (4).

Write the meridional entries of $\mathcal{P}(w\otimes w)$ in the cylindrical frame as $a = R_{rr}$, $b = R_{\theta\theta}$, $c = R_{zz}$, and $d = R_{rz}$. Their absolute values are at most K^2. Direct differentiation yields

$$F_r = -a_r - (a-b)/r - d_z, \qquad F_z = -d_r - d/r - c_z, \tag{5}$$

$$\frac{(\operatorname{curl} F)_\theta}{r} = \left(\partial_{rr} + \frac{3}{r}\partial_r - \partial_{zz}\right)\frac{d}{r} + \frac{1}{r}\partial_{rz}(c-a) - \frac{1}{r^2}\partial_z(a-b). \tag{6}$$

The quantities here are smooth at the axis in the appropriate lifted variables. For a distributional estimate it suffices that $d = O(r)$.

Consider the rescaling used in [1, Section 4]: for $t_n \uparrow 0$,

$$\tau_n = -t_n, \quad \lambda_n = \tau_n^{1/2}, \quad \delta_n = \tau_n^h, \quad \mu_n = \lambda_n/\delta_n, \quad r = \lambda_n R, \quad z = z_n + \mu_n Z, \quad t = \tau_n s.$$

The mean components are normalized as $V_n = \lambda_n v_r$, $W_n = \lambda_n\delta_n v_z$, $S_n = \lambda_n\delta_n v_\theta$, and the potential vorticity as $\Omega_n = \lambda_n^3\delta_n(\operatorname{curl} v)_\theta/r$. For a stress component a, let a_n denote its composition with this change of variables, without an amplitude normalization.

Lift $R = |X|$ to four radial dimensions. Since

$$\frac{1}{R}\partial_R H = \operatorname{div}_X\left(\frac{XH}{R^2}\right) - \frac{2H}{R^2},$$

the rescaled forcing in the potential-vorticity equation is

$$\begin{aligned} G_n = \lambda_n^2\delta_n\Big[&(\Delta_X - \delta_n^2\partial_{ZZ})(d_n/R) \\ &+ \delta_n\partial_Z\left\{\operatorname{div}_X\frac{X(c_n-a_n)}{R^2} - \frac{2(c_n-a_n)+(a_n-b_n)}{R^2}\right\}\Big]. \end{aligned} \tag{7}$$

Both R^{-1} and R^{-2} are locally integrable in $\mathbb{R}^4$. The divergence identity has no axis-supported mass: the flux through $R = \epsilon$ is $O(\epsilon^2)$ for bounded H. Move all derivatives in (7) onto a compactly

supported smooth test function with time support $J \Subset (-\infty, 0)$. The resulting absolute value is bounded by

$$C_J \lambda_n^2 \delta_n \int_J K(\tau_n s)^2 \, \mathrm{d}s = C_J \delta_n \int_{\tau_n J} K(t)^2 \, \mathrm{d}t \longrightarrow 0. \tag{8}$$

Thus no derivative of the fluctuation is needed in this part of the limiting argument. In a recentered zoom with $r = \lambda_n(A_n + R)$ and $A_n \to \infty$, the azimuthal-vorticity force has the same prefactor $\lambda_n^2 \delta_n$ multiplying at most two derivatives of the stress; its additional coefficients $(A_n + R)^{-1}$ and $(A_n + R)^{-2}$ are uniformly bounded on compact sets. Estimate (8) therefore applies there too.

3 Eliminating the ancient swirl

Lemma 3. *Let S on $[0, \infty) \times \mathbb{R} \times (-\infty, -1]$ have uniform spatial $W^{2,\infty}$ bounds on compact time intervals, be continuous through the upper time, and satisfy $S(0, Z, s) = 0$. The drifts V, W may be only measurable in time, but are uniformly bounded and spatially Lipschitz on each such interval. Suppose, for $R > 0$,*

$$S_s + V S_R + W S_Z + (V/R) S = S_{RR} + R^{-1} S_R - R^{-2} S \tag{9}$$

in distributions. If $h > 0$ and

$$|S(R, Z, s)| \le C(-s)^{-1/2-h}, \qquad |V(R, Z, s)| \le C_r(-s)^{-1/2},$$

then $S = 0$.

Proof. Put $\Gamma = RS$. Its equation on $R > 0$ is

$$\mathcal{L}\Gamma := \Gamma_s + V\Gamma_R + W\Gamma_Z - \Gamma_{RR} + R^{-1}\Gamma_R = 0.$$

For $L > 1$ let $B_L = CL^{-1/2-h}$ and

$$\Phi_L(R, s) = B_L R + C_r B_L \int_{-L}^{s} (-q)^{-1/2} \, \mathrm{d}q.$$

It dominates $|\Gamma|$ at $s = -L$, and

$$\mathcal{L}\Phi_L = B_L\big[C_r(-s)^{-1/2} + V + R^{-1}\big] \ge 0.$$

Comparison is valid on the unbounded spatial domain. On a fixed slab $[-L, s_1]$, perturb Φ_L by

$$\eta e^{k(s+L)}(1 + R^2 + Z^2).$$

The radial diffusion of R^2 is zero for this operator, so k can be chosen in terms of the bounded drifts to make the added function a supersolution. Its quadratic growth dominates the linear growth of Γ at infinity. Near the axis, $\Gamma = O(R^2)$ uniformly on this slab, while $\Phi_L \ge B_L R$. Thus one can excise a sufficiently small axis cylinder, on whose boundary comparison already holds. On the remaining bounded region the positive-part energy argument applies: spatial Lipschitz bounds control the divergence of the drift, and no diffusion in Z is needed. The distributional equation and spatial second-derivative bounds give local bounded time derivatives, so the time traces used in this argument are legitimate. This also justifies the argument for time-measurable drifts, as in [1, Lemma 3.3]. Let $\eta \downarrow 0$ and apply the comparison to both signs of Γ. For a fixed (R, Z, s) it yields

$$|\Gamma(R, Z, s)| \le CRL^{-1/2-h} + 2CC_r L^{-h} \longrightarrow 0 \qquad (L \longrightarrow \infty).$$

The upper time is included by continuity. □

To apply the lemma, first take a limit of the actual mean swirl equation. Write $e = R_{r\theta}$ and $f = R_{z\theta}$ for the corresponding entries of $\mathcal{P}(w \otimes w)$. They obey $|e| + |f| \le 2K^2$, and

$$F_\theta = -\partial_r e - 2e/r - \partial_z f.$$

In the preceding rescaling the swirl equation is

$$\begin{aligned}(S_n)_s + V_n(S_n)_R + W_n(S_n)_Z + (V_n/R)S_n = (S_n)_{RR} + R^{-1}(S_n)_R - R^{-2}S_n + \delta_n^2(S_n)_{ZZ} + H_n,\\ H_n = -\lambda_n^2\delta_n[\partial_R e_n + 2e_n/R + \delta_n\partial_Z f_n].\end{aligned} \tag{10}$$

For a rectangle $D \Subset \{R > 0\}$, a closed time slab $J = [-T, -1]$, and finite p, this gives

$$\|H_n\|_{L^1(J;W^{-1,p}(D))} \le C_{D,J}\delta_n \int_{\tau_n J} K(t)^2 \,\mathrm{d}t \longrightarrow 0. \tag{11}$$

The mean hypotheses uniformly bound V_n, W_n, S_n in spatial C^2 on this slab. All other terms of (10) are bounded in a negative Sobolev norm. Thus the time increments of S_n in that norm are bounded by $C|s - s'| + \epsilon_n$, where $\epsilon_n \to 0$. Interpolation with the spatial bounds gives asymptotic equicontinuity in C^0; spatial interpolation then gives locally uniform convergence of the first spatial derivatives too. Finitely many initial members are smooth, so this yields the usual compactness after a subsequence. The selected time $s = -1$ is included. The bounds $S_n(0, Z, s) = 0$ and $\|\partial_R S_n\|_\infty \le C_J$ extend the convergence to the axis. Also $\partial_Z S_n(0, Z, s) = 0$ and the mixed second derivative is uniformly bounded.

The drifts have weak-* limits, with the same global spatial Lipschitz bounds on each compact time slab. The strong convergence of S_n and its first derivatives permits passage to their products with these drifts. Equations (10)–(11) give (9) in the limit. The backward bounds are

$$|S(R, Z, s)| \le C|s|^{-1/2-h}, \qquad |V(R, Z, s)| \le C|s|^{-1/2}.$$

Their constants are independent of the compact spatial set. Lemma 3 therefore makes this limit zero.

For a recentered zoom with $r = \lambda_n(A_n + R)$ and $A_n \to \infty$, replace every curvature factor $1/R$ in (10) by $1/(A_n + R)$. The force estimate still holds on each fixed rectangle, and the limiting equation on $\mathbb{R}^2$ is simply

$$S_s + VS_R + WS_Z = S_{RR}.$$

Its bounded solution tends uniformly to zero in the remote past, so [1, Lemma 3.3] makes it zero as well.

In particular, on every smaller original ball,

$$(-t)\,\|\partial_z v_\theta(t)\|_\infty \longrightarrow 0. \tag{12}$$

Otherwise select points and times where this fails and use the same finite-axis or receding-axis alternatives. At the selected point $\partial_Z S_n = \lambda_n^2 \partial_z v_\theta$ tends to zero. If its scaled radial coordinate tends to zero, the conclusion instead follows directly from $\partial_Z S_n(0, Z, s) = 0$ and the mixed second-derivative bound. This proves (12).

4 Meridional smallness

Define, on a smaller ball,

$$G_v(t) = \| |\partial_r v_r| + |v_r/r| + |\partial_z v_z| + |\partial_z v_r| \|_\infty .$$

We verify the two changes needed in the proof of [1, Proposition 1.5] to conclude

$$(-t) G_v(t) \longrightarrow 0. \tag{13}$$

First, the preceding swirl argument gives $S_n \to 0$ on compact rescaled time slabs. The mean bounds also give $|S_n/R| \le C$ on those slabs, including the axis. Splitting at $R = \epsilon$ shows that S_n^2/R^2 tends to zero locally in L^1 in the lifted space. In the receding-axis case the source is $\partial_Z(S_n^2/(A_n + R))$, which likewise tends to zero in distributions.

Second, the force is small in a negative Sobolev norm integrated in time, rather than uniformly small as a function. Away from the axis, (7) and (8) give

$$\|G_n\|_{L^1(J;W^{-2,p}(D))} \longrightarrow 0$$

for a fixed rectangle D and any finite p. The analogous assertion holds in the receding-axis case. The other terms in the rescaled vorticity equation are uniformly bounded in this negative norm. Together with the uniform spatial Lipschitz bound supplied by (1)–(2), this gives an increment bound $C|s - s'| + \epsilon_n$ with $\epsilon_n \to 0$. Interpolation gives asymptotic uniform time equicontinuity, also at the selected upper time. Finitely many initial members are smooth, so Arzelà–Ascoli still applies. Boundedness and the small volume of $R < \epsilon$ give convergence across the axis.

The remaining ancient-limit argument is unchanged: the force and swirl source vanish, and the mean bounds supply the backward decay and bounded Lipschitz drift required by [1, Lemma 3.3]. The limiting vorticity vanishes. At the selected time, incompressibility and boundedness of the limiting meridional velocity then contradict a nonzero selected derivative, in either axis case. This proves (13). No bound on $\partial_r v_z$ is asserted.

5 Full-velocity enstrophy

Choose a radial cutoff χ whose derivative is supported in a regular annulus, and write

$$Y = \|\chi \operatorname{curl} u\|_2^2 , \quad M = \|\chi \nabla \operatorname{curl} u\|_2^2 , \quad A_v = \|\partial_z v_\theta\|_\infty .$$

Here A_v and G_v are taken on a ball containing the support of χ, with its boundary in the regular annulus; K is still taken on $B(1)$. All cutoff errors are uniformly bounded. Weighted L^2 contraction of angular averaging, including its tensor action on gradients, gives $Y_v \le Y$ and $M_v \le M$. Local div–curl estimates consequently bound the first derivatives of u, v by $C(Y+1)^{1/2}$ and their second derivatives by $C(M+1)^{1/2}$ in these weighted norms.

For $\Pi_\chi(q) = \int \chi^2 (\operatorname{curl} q \cdot \nabla) q \cdot \operatorname{curl} q$, put $\omega_q = \operatorname{curl} q$. There is an exact telescoping identity

$$\begin{aligned} \Pi_\chi(u) - \Pi_\chi(v) = \int \chi^2 \big[&(\omega_w \cdot \nabla) u \cdot \omega_u + (\omega_v \cdot \nabla) w \cdot \omega_u \\ &+ (\omega_v \cdot \nabla) v \cdot \omega_w \big]. \end{aligned} \tag{14}$$

Integrate the derivative on w off that factor in each term. The remaining factors have derivative orders one and two, apart from bounded cutoff errors. Hence, for every $\epsilon > 0$,

$$\begin{aligned} |\Pi_\chi(u) - \Pi_\chi(v)| &\le CK(Y+1)^{1/2}(M+1)^{1/2} + C \\ &\le \epsilon M + C_\epsilon (K^2+1)(Y+1). \end{aligned}$$

For the mean's stretching, write $a = (\omega_v)_r = -\partial_z v_\theta$, $b = (\omega_v)_\theta = \partial_z v_r - \partial_r v_z$, and $c = (\omega_v)_z = \partial_r v_\theta + v_\theta/r$. The cylindrical gradient identity underlying [1, Equation (5.7)] can be rewritten exactly as

$$(\omega_v \cdot \nabla)v \cdot \omega_v = a^2 \partial_r v_r + b^2 v_r/r + c^2 \partial_z v_z + 2ac\partial_z v_r - abc - 2ab\, v_\theta/r. \tag{15}$$

In particular the large derivative $\partial_r v_z$ has been replaced using the vorticity relation, and the last two terms have the coefficient a, whose supremum is A_v.

A local radial integration by parts gives

$$\|\chi v_\theta/r\|_2^2 \le \|\chi(\omega_v)_z\|_2^2 + C. \tag{16}$$

Indeed, after expanding $(\partial_r v_\theta + v_\theta/r)^2 r$, its cross term is $\partial_r(v_\theta^2)$. The axis boundary term vanishes because $v_\theta = O(r)$; the cutoff error is $2\int \chi \partial_r \chi\, v_\theta^2 \,\mathrm{d}r\,\mathrm{d}z\,\mathrm{d}\theta$. It is bounded on the regular annulus, including its intersection with the axis, where $\partial_r \chi/r$ is bounded. Applying Cauchy–Schwarz to the last two terms of (15), and using (16), yields

$$\Pi_\chi(v) \le C(G_v + A_v)(Y_v + 1).$$

Testing the original unforced vorticity equation and absorbing the small dissipation term from the full-velocity difference now gives

$$Y' + cM \le C_*[G_v + A_v + K^2 + 1](Y + 1). \tag{17}$$

Fix $\eta > 0$. Equations (13) and (12) allow a final interval on which $G_v + A_v \le \eta/\tau$. Since $K \in L_t^2$, Gronwall in (17) gives

$$Y(t) + 1 \le C_\eta(-t)^{-C_*\eta}.$$

Here C_* was fixed before η. Choose η with $2C_*\eta < 1$. Local Sobolev now places u in $L_t^4 L_x^6$ on a smaller cylinder. The local criterion of [2, Theorem 1.1(i)], applied on backward cylinders ending at zero, makes $(0,0)$ regular. This finishes the proof of Theorem 1.

6 A separate quantitative gradient-history estimate

Regularity in Theorem 1 requires only (3). For applications which additionally require an absolute gradient or torque history, the actual defect equation gives the following stronger estimate. The whole-space assumptions here control the pressure produced by localization; they are not assumptions of Theorem 1.

Theorem 4. *Let u be a smooth unforced solution on $\mathbb{R}^3 \times (-1,0)$ with $u \in L^\infty((-1,0);L^2(\mathbb{R}^3))$ and canonical pressure, whose restriction to Q satisfies the local suitability assumptions above. For some $0 < h < 1/2$, assume only the following mean bounds on $B(1)$:*

$$\|v(-\tau)\|_\infty \le C_M \tau^{-1/2-h}, \qquad \|\nabla v(-\tau)\|_\infty \le C_N \tau^{-1-h}, \tag{18}$$

with Cartesian derivatives. Suppose, for some $0 < a < 1$,

$$\int_0^a \tau^{-h/2}\sqrt{\log(e/\tau)}\, \|w(-\tau)\|_{L^\infty(B(1))}^2 \,\mathrm{d}\tau < \infty. \tag{19}$$

Then, on a smaller ball and final time interval,

$$\int \sqrt{-t}\, \|w(t)\|_\infty \|\nabla w(t)\|_\infty \,\mathrm{d}t < \infty. \tag{20}$$

It also bounds the same weighted integral of $\|(\mathcal{P}[(w\cdot\nabla)w])_\theta\|_\infty$. *For this additional conclusion, the amplitude power* $\|w(-\tau)\|_\infty \le C\tau^{-\beta}$ *suffices when* $\beta < 1/2 - h/4$. *At equality, a factor* $\log(e/\tau)^{-p}$ *suffices for* $p > 3/4$. *If the full mean geometry of Theorem 1 also holds, regularity already follows from the weaker condition (3).*

The theorem does not assert (19) from initial data. Its proof has a whole-space smoothing estimate and a localization step. On a sufficiently short final interval it bounds (20) by a constant times the integral in (19), plus a bounded annular-source contribution. Write $\mathbb{L}$ for the Leray projection, to distinguish it from angular averaging $\mathcal{P}$.

Lemma 5. *Let a smooth divergence-free* U *on* $\mathbb{R}^3\times[0,d]$ *satisfy*

$$U_t - \Delta U = -\mathbb{L}\,\mathrm{div}(U\otimes U) + g, \qquad V = \mathcal{P}U, \quad W = U - V, \quad G = (1-\mathcal{P})g.$$

Assume the fields and the spatial derivatives used below are bounded on compact time prefixes, and that $\|V\|_\infty \le M$, $\|\nabla V\|_\infty \le N$, $\|G\|_\infty \le G_0$ *there, with* $M, N > 0$. *Put* $K(q) = \|W(q)\|_\infty$, $L(q) = \|\nabla W(q)\|_\infty$, *and*

$$Z_d = \sup_{0<q\le d}\{K(q) + \sqrt{q}L(q)\}, \qquad \log_+ x = \max(\log x, 0).$$

There are universal constants such that

$$Z_d \le C_0(K(0) + dG_0) + C_1 N d\left[1 + \log_+\frac{M}{N\sqrt{d}}\right] Z_d + C_2\sqrt{d}Z_d^2. \tag{21}$$

Consequently $Z_d \le 2C_0(K(0)+dG_0)$ *if the linear coefficient is at most* $1/4$ *and* $\sqrt{d}(K(0)+dG_0)$ *is sufficiently small.*

Proof. Rotation covariance gives the exact homogeneous-in-W equation

$$W_t - \Delta W = -\mathbb{L}\,\mathrm{div}\left[V\otimes W + W\otimes V + (1-\mathcal{P})(W\otimes W)\right] + G. \tag{22}$$

In particular the complete defect self-interaction is retained. For the heat kernel H_s, the kernel of $e^{s\Delta}\mathbb{L}\,\mathrm{div}$ is

$$(B_s)_{ijk} = \delta_{ij}\partial_k H_s + \int_s^\infty \partial_{ijk}H_b\,\mathrm{d}b.$$

Differentiating the Gaussian and integrating in b gives

$$|B_s(y)| \le C(\sqrt{s}+|y|)^{-4}, \qquad |\nabla B_s(y)| \le C(\sqrt{s}+|y|)^{-5},$$
$$\|B_s\|_1 \le Cs^{-1/2}, \qquad \int |y|\,|\nabla B_s(y)|\,\mathrm{d}y \le Cs^{-1/2}.$$

The first absolute moment of B_s is not integrable at infinity. Splitting the radial integral at $\sqrt{s}$ and M/N instead gives

$$\int |B_s(y)|\min(N|y|, 2M)\,\mathrm{d}y \le CN\left[1 + \log_+\frac{M}{N\sqrt{s}}\right], \tag{23}$$
$$\int |\nabla B_s(y)|\min(N|y|, 2M)\,\mathrm{d}y \le CNs^{-1/2}. \tag{24}$$

For each endpoint (x,q) follow the backward characteristic $z'(s) = V(z(s),s)$, $z(q) = x$, and set $a(s) = V(z(s),s)$. Move $a(s)\cdot\nabla W$ to the left side of (22); its two linear tensors now contain $V - a$

instead of V, since $\operatorname{div} W = 0$. This identity holds everywhere with this one spatially constant $a(s)$. The translated Duhamel kernels at the chosen endpoint are centered at $z(s)$, where

$$|V(y,s) - a(s)| \le \min(N|y - z(s)|, 2M).$$

For the gradient estimate, differentiate that representation with $a(s)$ held fixed. No derivative of the selected characteristic is introduced: the same representation is valid at every nearby endpoint with this fixed a. Estimates (23)–(24) bound the linear amplitude and gradient terms.

For the quadratic gradient term move the derivative onto the tensor: $\|\nabla(1-\mathcal{P})(W\otimes W)\|_\infty \le CKL$. Angular averaging and its tensor action are contractions in the Euclidean sup norms. The time integral $\int_0^q (q-s)^{-1/2}s^{-1/2}\,\mathrm{d}s = \pi$ then bounds this term by CZ_d^2, before multiplication by $\sqrt{q}$. The quadratic amplitude term is bounded by $C\sqrt{q}Z_d^2$. The initial term and the divergence-free source use the ordinary translated heat kernel. Finally

$$\int_0^q \left[1 + \log_+ \frac{M}{N\sqrt{s}}\right] \mathrm{d}s \le Cq\left[1 + \log_+ \frac{M}{N\sqrt{q}}\right].$$

This proves (21). The estimate holds on every shorter prefix, whose linear coefficient is no larger. A continuity bootstrap gives the final claim; smoothness supplies the initial continuity. If $K(0) + dG_0 = 0$, use uniqueness or a positive bootstrap bound tending to zero. □

Proof of Theorem 4. First suppose a globally bounded defect equation as in Lemma 5 has been obtained, with G uniformly bounded and

$$\|V(-\tau)\|_\infty \le C_M \tau^{-1/2-h}, \qquad \|\nabla V(-\tau)\|_\infty \le C_N \tau^{-1-h}. \tag{25}$$

Let $\ell(\tau) = \log(e/\tau)$ and choose

$$d(\tau) = \frac{c\tau^{1+h}}{\ell(\tau)}, \qquad 0 < c < 1. \tag{26}$$

On $[-\tau - d, -\tau]$, the linear coefficient in (21) is at most $Cc(1 + |\log c| + h)$: here $Nd \le C_N c/\ell$ and $M/(N\sqrt{d})$ is bounded by a constant times $c^{-1/2}\tau^{-h/2}\sqrt{\ell}$. Take c small enough in terms of the mean bounds.

Write $K = \|W\|_\infty$, $L = \|\nabla W\|_\infty$, and

$$E(\tau) = \int_\tau^{\tau + d(\tau)} K(-s)^2\,\mathrm{d}s.$$

Condition (19), with the localized global norm, implies $K \in L_t^2$ and $E(\tau) \to 0$. There is a time $s_0 \in [-\tau - d, -\tau - d/2]$ at which $K(s_0)^2 \le 2E(\tau)/d$. Applying Lemma 5 on $[s_0, -\tau]$ is therefore legitimate for all sufficiently small τ; its small parameter is at most $C\sqrt{E(\tau)} + d^{3/2}G_0 \to 0$. We obtain

$$K(-\tau) + \sqrt{d}L(-\tau) \le C[d^{-1/2}\sqrt{E(\tau)} + dG_0]. \tag{27}$$

Consequently

$$\sqrt{\tau}K(-\tau)L(-\tau) \le C\sqrt{\tau}d^{-3/2}E(\tau) + C\sqrt{\tau}d^{3/2}G_0^2. \tag{28}$$

The second term is integrable. For the first, Tonelli gives

$$\int_0^{a/2} \sqrt{\tau}d(\tau)^{-3/2}E(\tau)\,\mathrm{d}\tau \le C\int_0^a s^{-h/2}\sqrt{\ell(s)}K(-s)^2\,\mathrm{d}s. \tag{29}$$

Indeed, $\tau \le s \le \tau + d(\tau)$ implies $s/(1+c) \le \tau \le s$. Since d increases, these τ lie in an interval of length at most $d(s)$, while the integrand's coefficient is at most $Cs^{-1-3h/2}\ell(s)^{3/2}$. This proves (29), hence the gradient history on a final interval. The angularly averaged torque is at most KL.

It remains to construct the localization used above. Choose a radial cutoff χ, equal to one on an interior ball and with its transition inside a regular annulus of the original solution. Put

$$\Psi = (-\Delta)^{-1} \operatorname{curl} u, \qquad U = \operatorname{curl}(\chi\Psi) = \chi u + b, \qquad b = \nabla\chi \times \Psi.$$

The L^2 hypothesis and incompressibility give $\operatorname{curl}\Psi = u$; the potential is well defined by its order-minus-one kernel. On the annulus, its separated kernel and every required derivative are bounded against $\|u\|_2$, while its local part is controlled by interior spatial regularity. For canonical pressure, the same decomposition for its order-zero transform of $u \otimes u$ bounds p and its spatial derivatives on the annulus. Likewise

$$\Psi_t = -\operatorname{curl} u - (-\Delta)^{-1} \operatorname{curl}\operatorname{div}(u \otimes u)$$

has bounded spatial derivatives there: its bilinear operator is of order zero, with separated kernels controlled by $\|u\|_2^2$. These statements use only annular regularity and the uniform energy bound, not regularity of the interior through time zero.

With pressure χp, the exact force of U is

$$\begin{aligned} f_U = b_t - 2\nabla\chi\cdot\nabla u - (\Delta\chi)u - \Delta b + p\nabla\chi \\ + (U\cdot\nabla)U - \chi(u\cdot\nabla)u. \end{aligned} \tag{30}$$

It is supported in the regular annulus and uniformly bounded in spatial C^1. Its Leray projection is uniformly bounded, as follows by subtracting $f_U(x)$ in the local principal value and using its L^1 norm for the far part. Thus the equation for U has $g = \mathbb{L}f_U$ and bounded $G = (1-\mathcal{P})g$, including pressure.

Rotations preserve the interior ball, so $U = u$, $\mathcal{P}U = v$, and $U - \mathcal{P}U = w$ there. The transition region contributes bounded fields and derivatives; outside the cutoff $U = 0$. The mean assumptions therefore imply (25) globally. The global weighted amplitude integral is bounded by (19) plus the finite contribution of a bounded function. Equations (27)–(29) now supply (20) on a smaller cylinder of the original flow. The power and logarithmic consequences follow by integrating $\tau^{-h/2-2\beta}\ell(\tau)^{1/2-2p}$. □

7 The sign of kinetic mean–defect transfer

For a smooth finite-energy whole-space unforced flow, orthogonality of angular averaging and integration by parts give

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|w\|_2^2 + \nu\|\nabla w\|_2^2 = -T, \qquad \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|v\|_2^2 + \nu\|\nabla v\|_2^2 = T, \quad T = \int w\cdot(w\cdot\nabla)v. \tag{31}$$

We display viscosity explicitly in this section. Defect self-advection and pressure do no kinetic work on w; the averaging correction is orthogonal to w. In particular the signed net kinetic transfer obeys

$$-\int_0^t T(s)\,\mathrm{d}s = \tfrac{1}{2}\|v(0)\|_2^2 - \tfrac{1}{2}\|v(t)\|_2^2 - \nu\int_0^t \|\nabla v\|_2^2\,\mathrm{d}s \le \tfrac{1}{2}\|v(0)\|_2^2 .$$

This does not bound its positive variation or the amplitude integral (19). The following example also excludes an automatic viscous damping argument based just on small relative angular defect.

Proposition 6. *For every $\nu > 0$, $C > 0$, and $0 < \epsilon \le 1$, there are real Schwartz solenoidal data u_+ and u_- with $\|u_\pm\|_2 = 1$ and the same angular mean v. Their defects $w_\pm = u_\pm - v$ have equal squared L^2 and gradient norms, and $\|w_\pm\|_2 = (3\epsilon/2)\,\|v\|_2$. Their nonlinear kinetic transfers have opposite signs, and the unforced solution from u_+ satisfies at its initial time*

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\,\|w_+(t)\|_2^2 > C\nu\,\|\nabla w_+\|_2^2\,.$$

The mean satisfies the spatial bounds (1)–(2) at a time $t = -\tau_\lambda$, with $h = 1/6$ and a constant independent of the concentration parameter λ and of $\epsilon \in (0, 1]$. This last assertion concerns that time slice, not an entire evolution.

Proof. Set $\rho = x^2 + y^2 + z^2$, $r^2 = x^2 + y^2$, and, for $\sigma = \pm 1$,

$$V = \operatorname{curl}(e^{-\rho}e_z), \qquad W_\sigma = \operatorname{curl}\left((x^2 - y^2 + 2\sigma xyr^2)e^{-\rho}e_z\right).$$

Both fields are horizontal, divergence free and Schwartz; $\mathcal{P}V = V$ and $\mathcal{P}W_\sigma = 0$. Write $g_a = (\pi/a)^{3/2}$. Gaussian moments give

$$\begin{aligned} \|V\|_2^2 &= 2g_2, & \|W_\sigma\|_2^2 &= \tfrac{9}{2}g_2,\\ \|\nabla_h W_\sigma\|_2^2 &= 30g_2, & \|\partial_z W_\sigma\|_2^2 &= \tfrac{9}{2}g_2,\\ \int W_\sigma \cdot (W_\sigma \cdot \nabla)V &= -\sigma\tfrac{16}{9}g_3. \end{aligned}$$

For example, the radial–azimuthal entry of $\mathcal{P}(W_\sigma \otimes W_\sigma)$ is $2\sigma r^4 e^{-2\rho}$, while $\partial_r V_\theta - V_\theta/r = -4r^2e^{-\rho}$; their product directly gives the last integral.

Let $\mu = \lambda^{2/3}$, $\lambda \ge 1$, and put

$$C_\epsilon = g_2(2 + \tfrac{9}{2}\epsilon^2), \qquad A = \frac{\lambda\sqrt{\mu}}{\sqrt{C_\epsilon}}, \qquad u_\sigma(x, y, z) = A(V + \epsilon W_\sigma)(\lambda x, \lambda y, \mu z).$$

The absence of an axial component preserves incompressibility under this anisotropic scaling. Orthogonality proves the stated unit norm and relative defect norm. The positive nonlinear injection $I_+ = -T_+$ and its viscous cost νD_+ obey

$$\frac{I_+}{\nu D_+} = \frac{A\lambda(16g_3/9)}{\nu g_2(30\lambda^2 + \frac{9}{2}\mu^2)} \longrightarrow \infty.$$

Indeed this ratio grows as a positive constant times $\lambda^{1/3}/\nu$, with ϵ fixed. Choose it larger than $C + 1$ and apply (31). For $\sigma = -1$ the injection changes sign while the dissipation is unchanged. These are jets of actual local unforced solutions from Schwartz data; no imposed strain or truncated source is used. Continuity also gives a nonzero initial interval with the strict inequality, but no uniform length of that interval is asserted.

Finally take $\tau_\lambda = \lambda^{-2}$. The mean has only an azimuthal component, whose (a, b) derivative is bounded by

$$C_{ab}\lambda^{4/3+a+(2/3)b} = C_{ab}\tau_\lambda^{-1/2-1/6-a/2-(1/2-1/6)b}.$$

The constants are uniform for $0 < \epsilon \le 1$ because $C_\epsilon \ge 2g_2$. This proves the asserted spatial bounds. □

The example refutes zero-allowance viscous absorption of kinetic mean–defect transfer, even with small relative defect and the stated spatial geometry. It does not refute a data-dependent history allowance, control of repeated interactions, or Theorems 1 and 4. In particular these transfers are kinetic energy transfers, not vorticity stretching integrals.

There is also an exact circulation test. The initial mean circulation $\Gamma = rv_\theta$ has its global maximum at $r = 1/\lambda$, $z = 0$, with value $2A/(e\lambda)$. For the base Gaussian defect,

$$rF_\theta = 4\sigma r^4(2r^2 - 3)e^{-2\rho}, \qquad F = -\mathcal{P}[(W_\sigma \cdot \nabla)W_\sigma].$$

At that maximum the radial and axial diffusion of the base circulation are $-8/e$ and $-4/e$. Consequently the actual initial mean equation gives

$$\Gamma_t(1/\lambda, 0) = -\frac{4\nu A}{e\lambda}(2\lambda^2 + \mu^2) - \frac{4\sigma\epsilon^2 A^2}{e^2}. \tag{32}$$

For $\sigma = -1$, choose λ still larger so that $\epsilon^2 A/(\nu\lambda) > e[2 + (\mu/\lambda)^2]$. The value at the old maximum then increases initially, and hence exceeds the initial global maximum. Its energy comes from the defect in (31). Thus the maximum principle for exactly axisymmetric circulation cannot be extended by dropping the Reynolds torque, even for arbitrarily small relative angular defect. This is an initial growth example, not repeated spin-up or singularity formation.

8 Scope of the calculation

The theorem allows an unbounded angular fluctuation, but requires its history to satisfy (3) and separately assumes the mean geometry. Neither hypothesis is established here for general data. Theorem 4 separately supplies a quantitative gradient history under a stronger weighted amplitude condition. Its localization constants may depend on the solution on a regular annulus; they are not uniform data-only bounds. In Theorem 1, an original force uniformly bounded in spatial C^2 on Q can also be included: its rescaled swirl and vorticity sources vanish, and its enstrophy contribution is bounded by $C(Y + 1)$. Thus a distinction between forced and unforced flows must enter a proof of the hypotheses, not be inferred from this criterion.

The accompanying symbolic calculation checks the mean equation, cylindrical stress divergence, lifted force identity, and the stretching integration by parts on explicit polynomial fields with all six stress components nonzero. It also checks the general cylindrical stretching and local radial integration identities, and the Gaussian moments and complete convection work in Proposition 6. The polynomial fields are local algebra tests, whereas the Gaussian fields are finite-energy initial data. The program does not integrate solutions or certify the analytic proof. No statement in this note has been added as an axiom or a theorem to the Lean library.

References

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